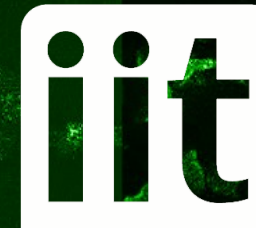




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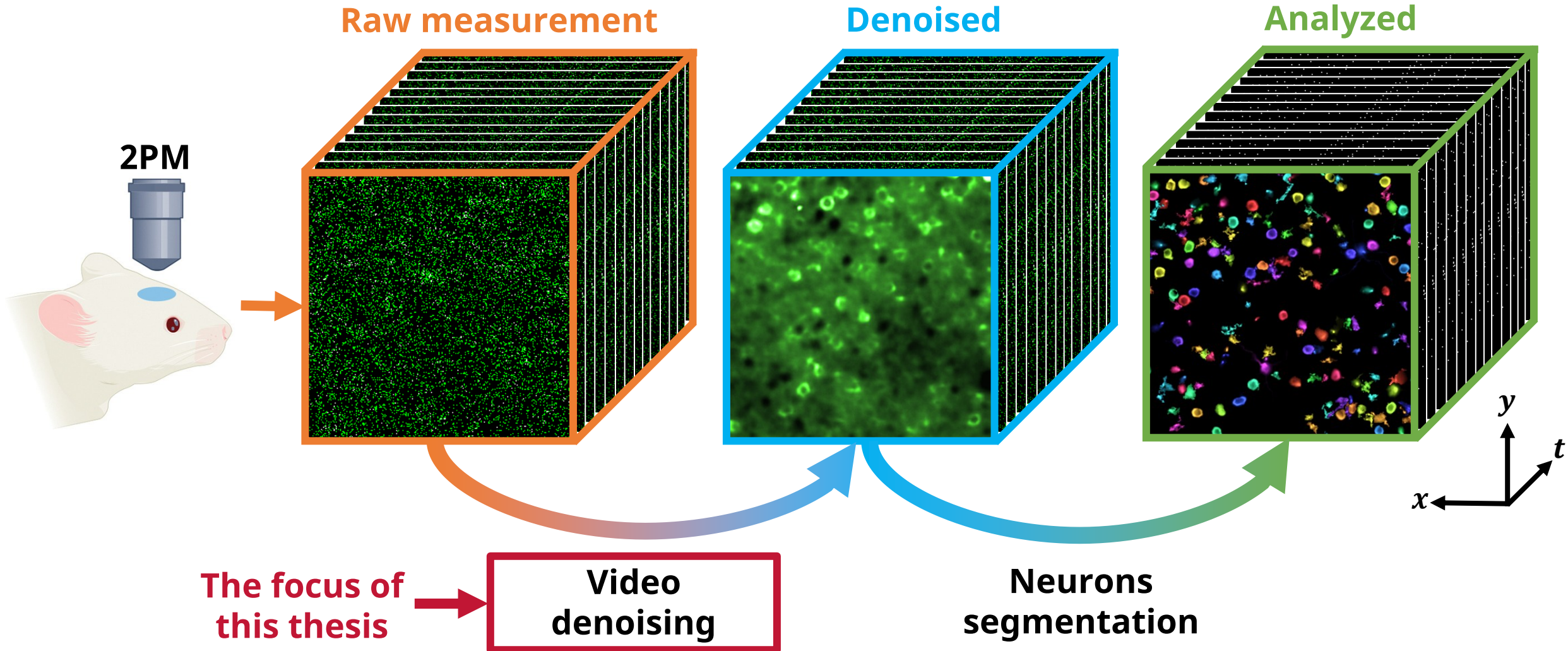
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Freq2Clean: enhancing calcium imaging denoising via frequency-domain video fusion

Master thesis presentation

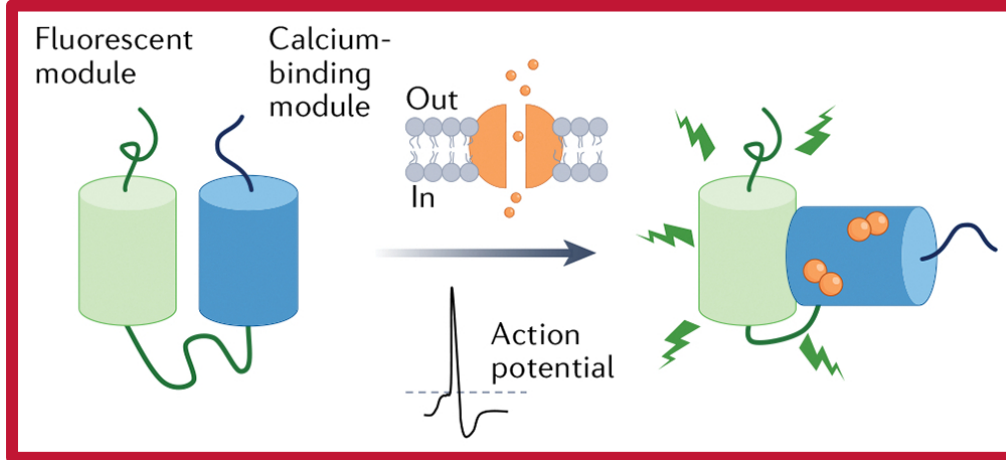
Candidate: **Valerio Morelli**
Tutor: **Prof. Adriano Mancini**
1st Co-tutor: **Prof. Vittorio Murino**
2nd Co-tutor: **Daniele Berardini, PhD**
October 17, 2025

The classic neuroimaging pipeline



Calcium imaging

Microscopy-based neuroimaging technique that indirectly **detects cellular activity by measuring changes in intracellular calcium concentration.**

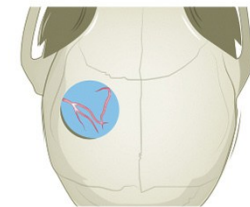


Ca^{2+} as a second messenger - When a neuron fires an action potential, calcium channels open, allowing Ca^{2+} to flow into the cytoplasm

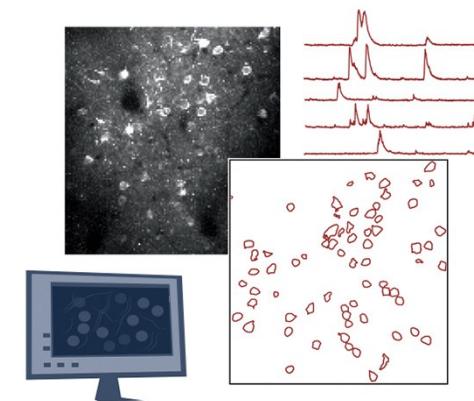
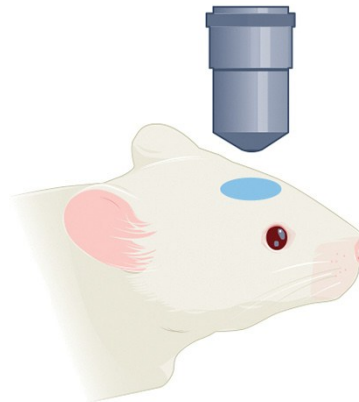
Fluorescent indicators - bind to Ca^{2+} and change their brightness property:



① GECI expression



② Cranial window and 2P calcium imaging



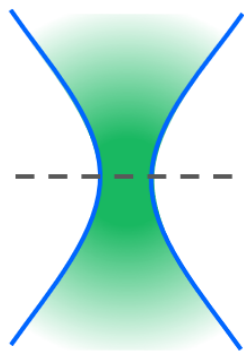
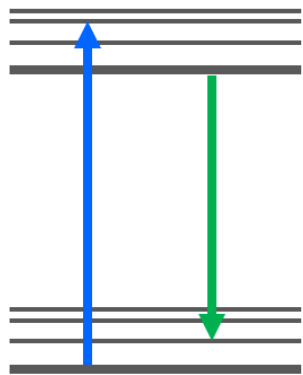
③ Signal analysis

- **Synthetic dyes** - Temporary, *strong* fluorescence;
- **GECIs** - Persistent, *weak* fluorescence.

2-Photon Microscopy

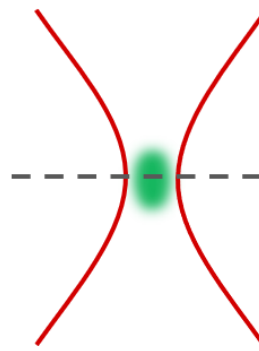
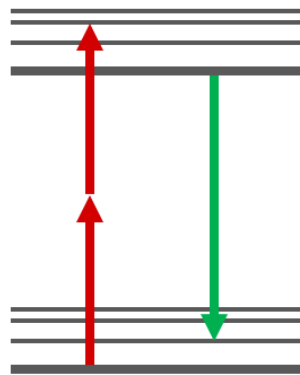
Two photons, of lower energy than needed for 1-photon excitation, can also excite a GFP in one quantum event. **Near-simultaneous absorption of two photons is highly improbable.**

1 Photon Excitation



$$I_F \propto I_{\text{Laser}}$$

2 Photon Excitation



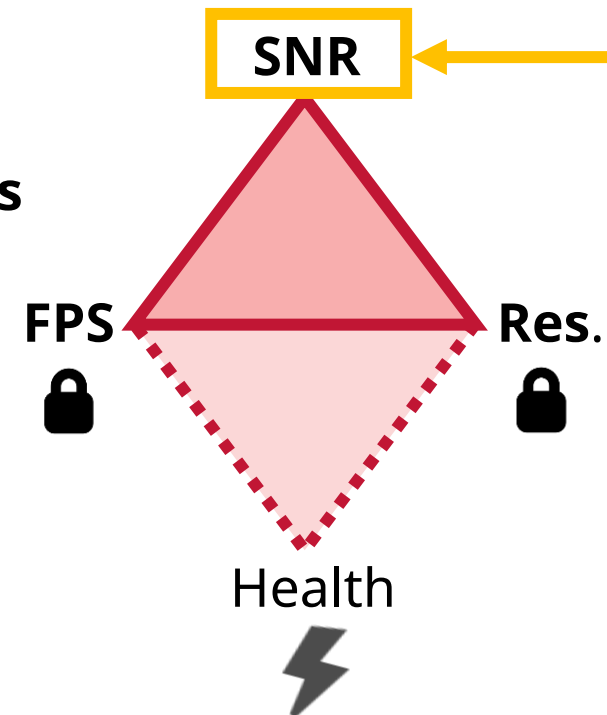
$$I_F \propto I_{\text{Laser}}^2$$

⇒ **Less stress** on tissue, **deeper** imaging.

However, there are still some **major limitations**:

- ❑ Low *photon budget* ⇒ **low SNR**
- ❑ Low *quantum yield* ⇒ **low SNR**
- ❑ *Photobleaching* ⇒ **short sessions**
- ❑ *Phototoxicity* ⇒ **low emission**
- ❑ *Fast spikes* ⇒ **low dwell time**
- ❑ *Deep imaging* ⇒ **scattering**
- ❑ *Shot noise* ⇒ **stochasticity**

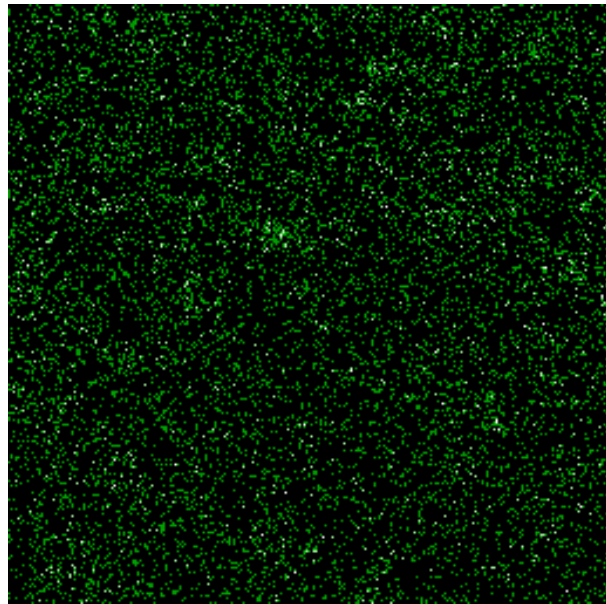
I_F = Excitation probability



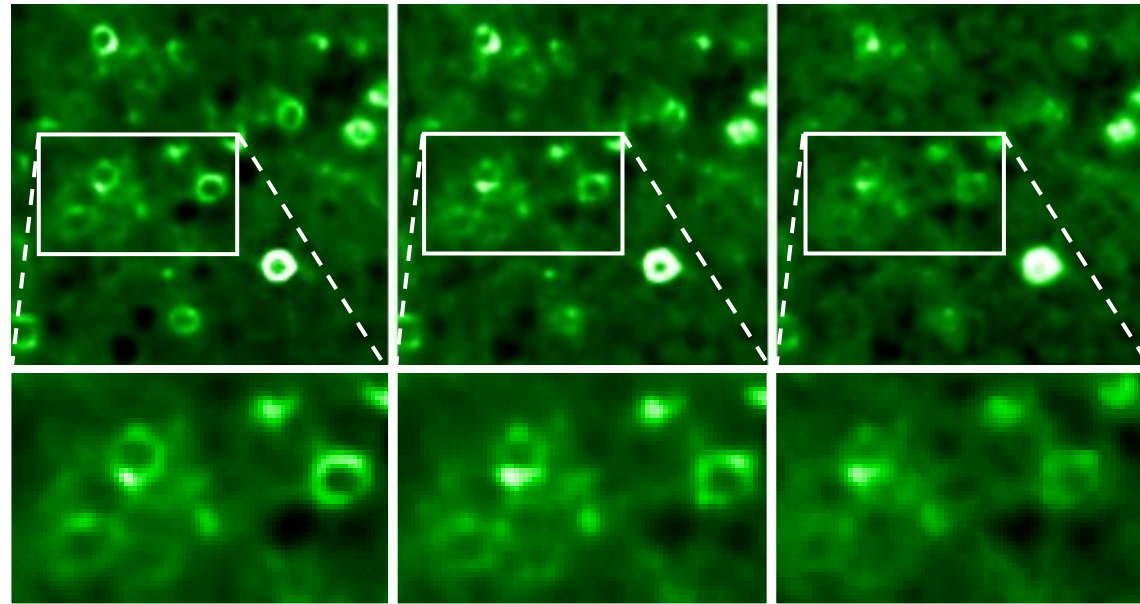
Aim of the thesis

Non-synthetic ground truths are scarce $\Rightarrow$ **Self supervised denoisers are preferable**

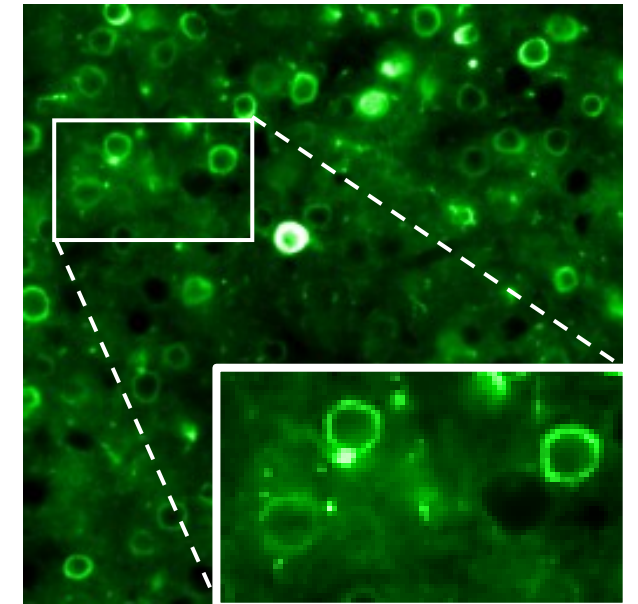
The main drawback of self-supervised methods is their inability to capture fine details, which results in images that are overly smooth.



Input



State-of-the-art denoisers

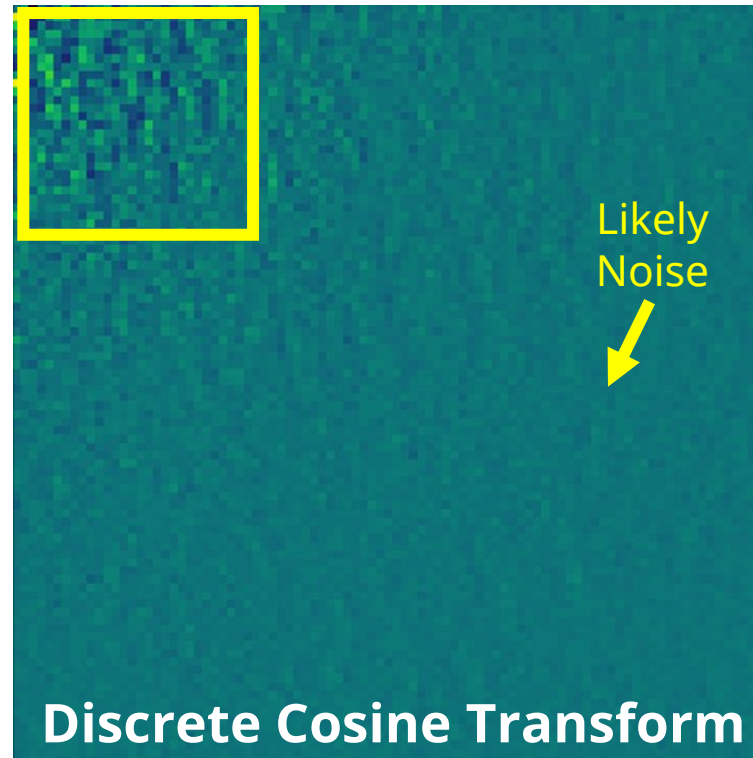
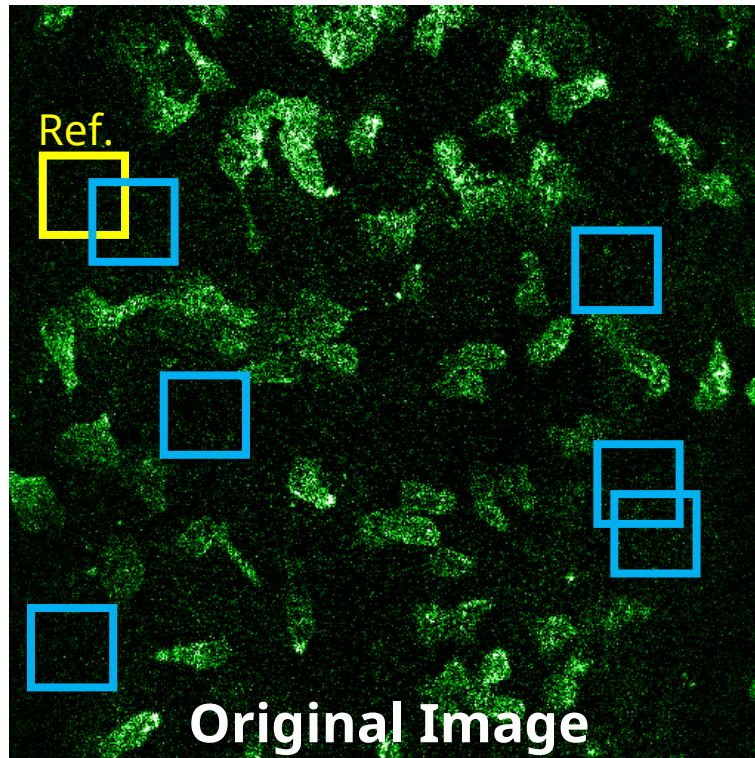


Ground Truth

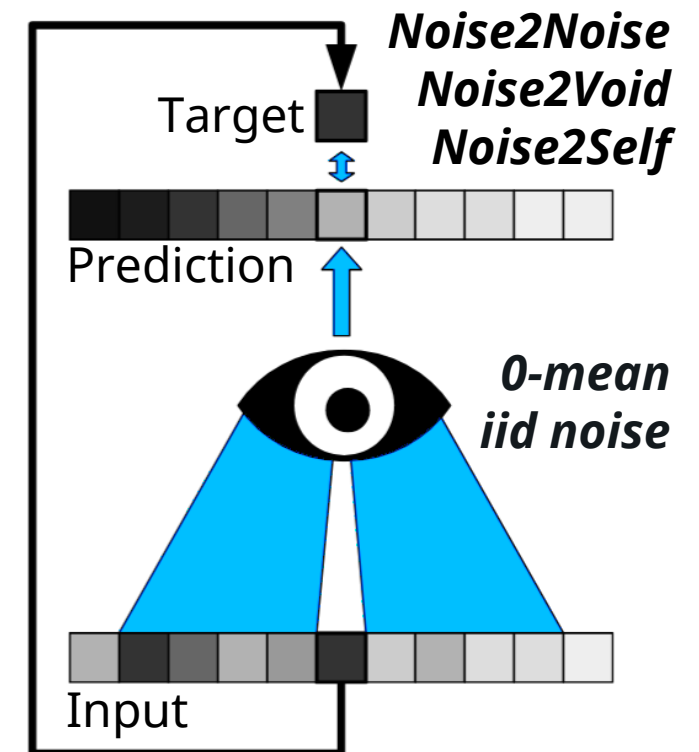
Principles of image denoising

Image denoising is an *ill-posed* problem. We need a way to **regularize** the denoising, among many possible solutions.

*Self-similarity,
Sparsity in a transform domain*



*Blind-spot
networks*



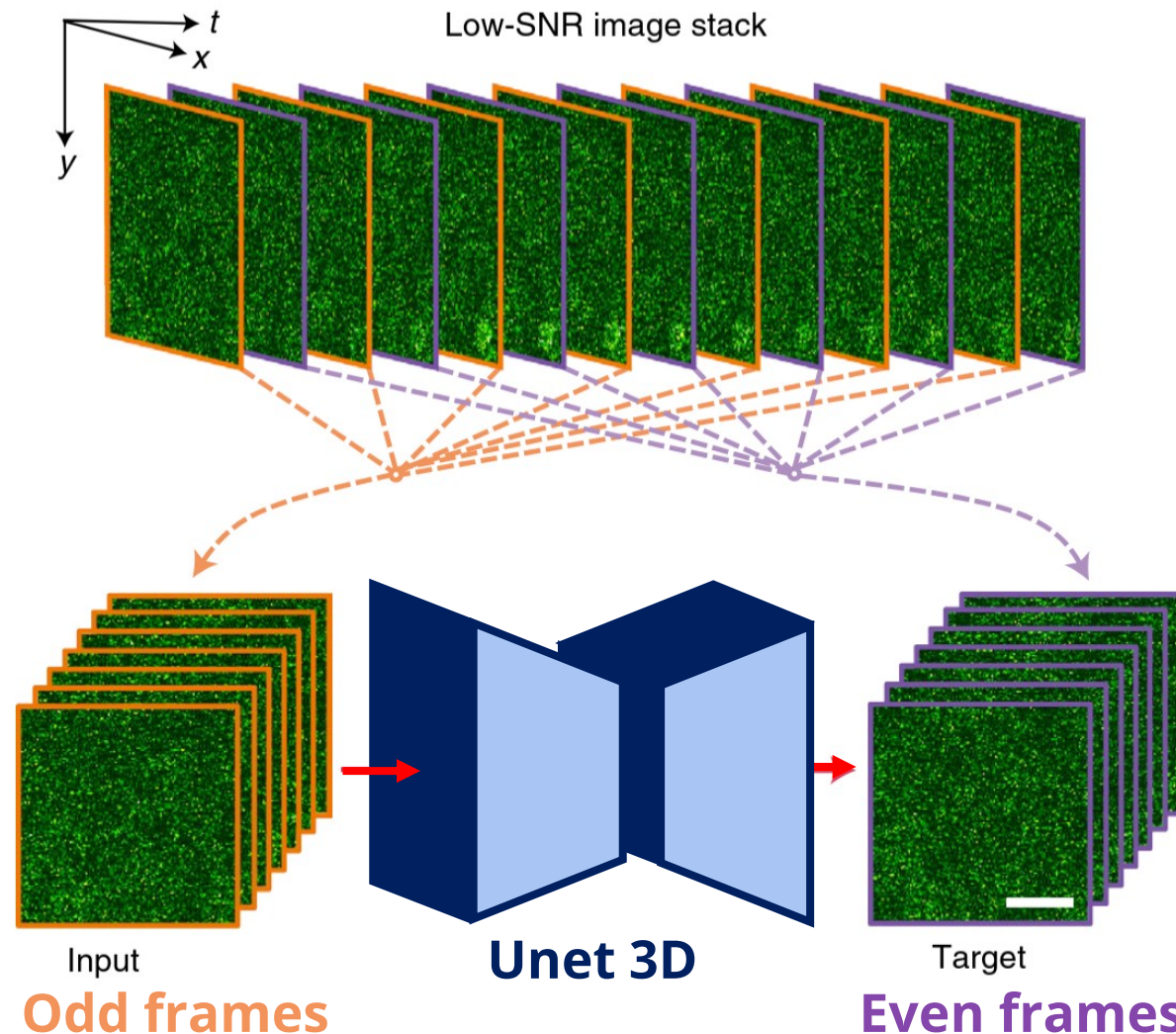
DeepCAD-RT: SOTA in 2PM denoising

Contextualizes *Noise2Noise* to 2PM:

- Hp1 → **0-mean** noise (*unbiased*)
- Hp2 → **temporally independent**

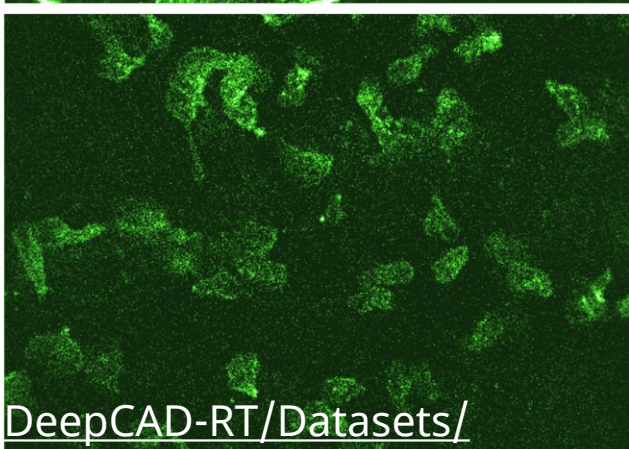
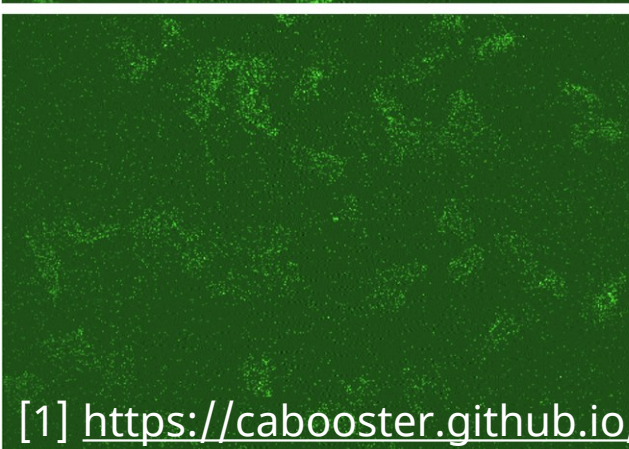
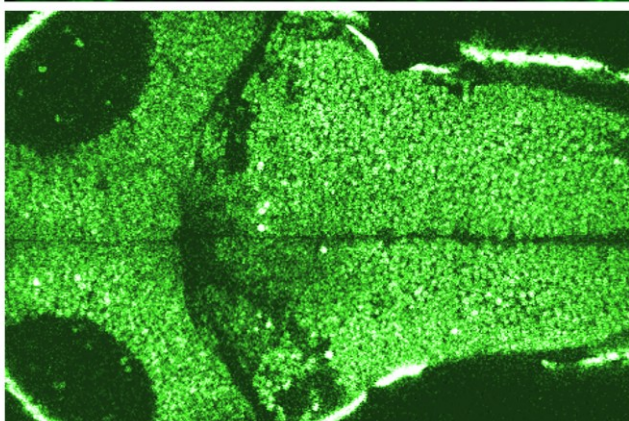
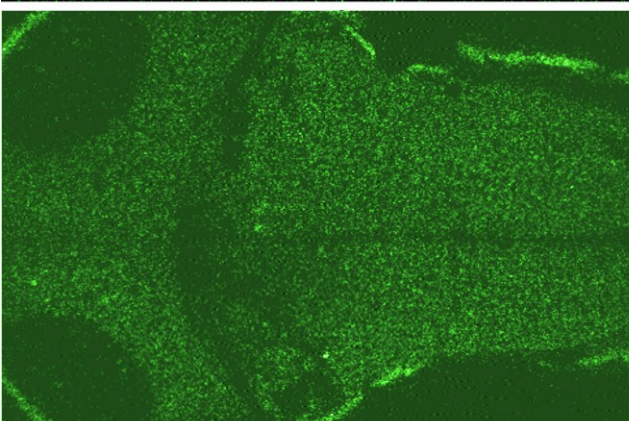
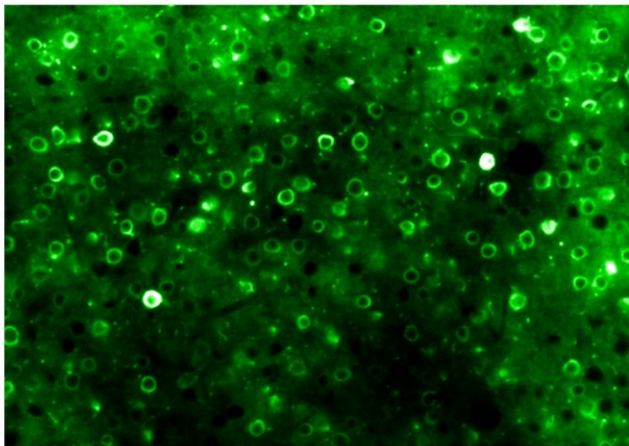
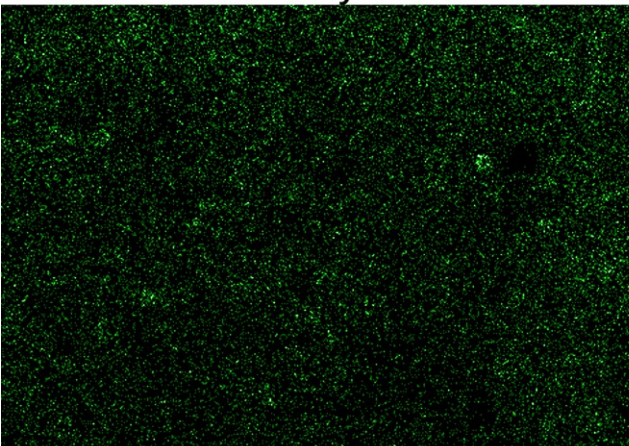
Trained to predict subsequent frames, the prediction converges to clean frames.

- ❑ Often trained to overfit on single recordings
- ❑ Fast training/inference, scale linearly with the number of frames considered in each patch



Noisy

Ground Truth



Material & methods

From DeepCAD repository [1], **three datasets** were selected:

- *Synthetic with ground truths* – the main reference for experiments
- Zebrafish brain – moderate input SNR
- *Real* mouse neutrophils – has spatial dynamics

For denoising evaluation:

↑ **PSNR** = $10 \log_{10} \left(\frac{\text{MAX}^2}{\frac{1}{n} \sum_{i=0}^n ||x_i - y_i||_2^2} \right)$

↑ **SSIM** = $\left(\frac{2\mu_x\mu_y + \epsilon_1}{\mu_x^2 + \mu_y^2 + \epsilon_1} \right)^\alpha + \left(\frac{2\sigma_x\sigma_y + \epsilon_2}{\sigma_x^2 + \sigma_y^2 + \epsilon_2} \right)^\beta + \left(\frac{\sigma_{x,y} + \epsilon_3}{\sigma_x\sigma_y + \epsilon_3} \right)^\gamma$

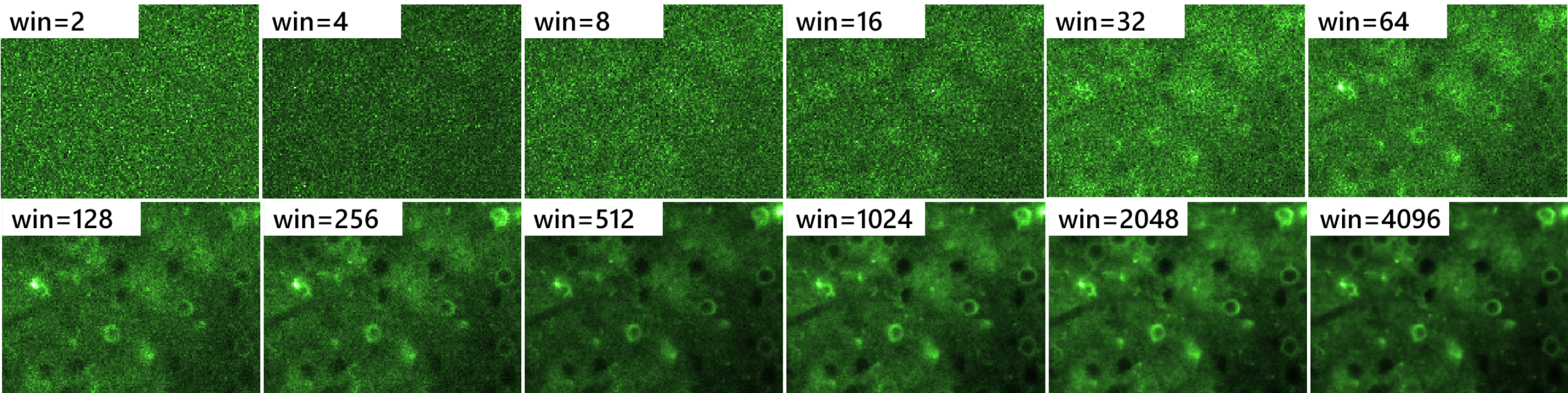
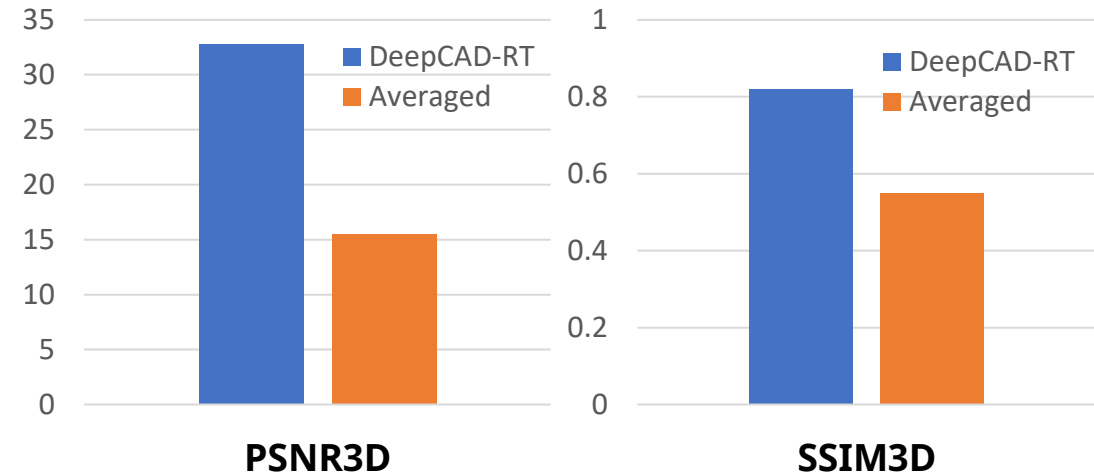
luminance **contrast** **structure**

[1] <https://cabooter.github.io/DeepCAD-RT/Datasets/>

Averaging increases SNR but smooths dynamics

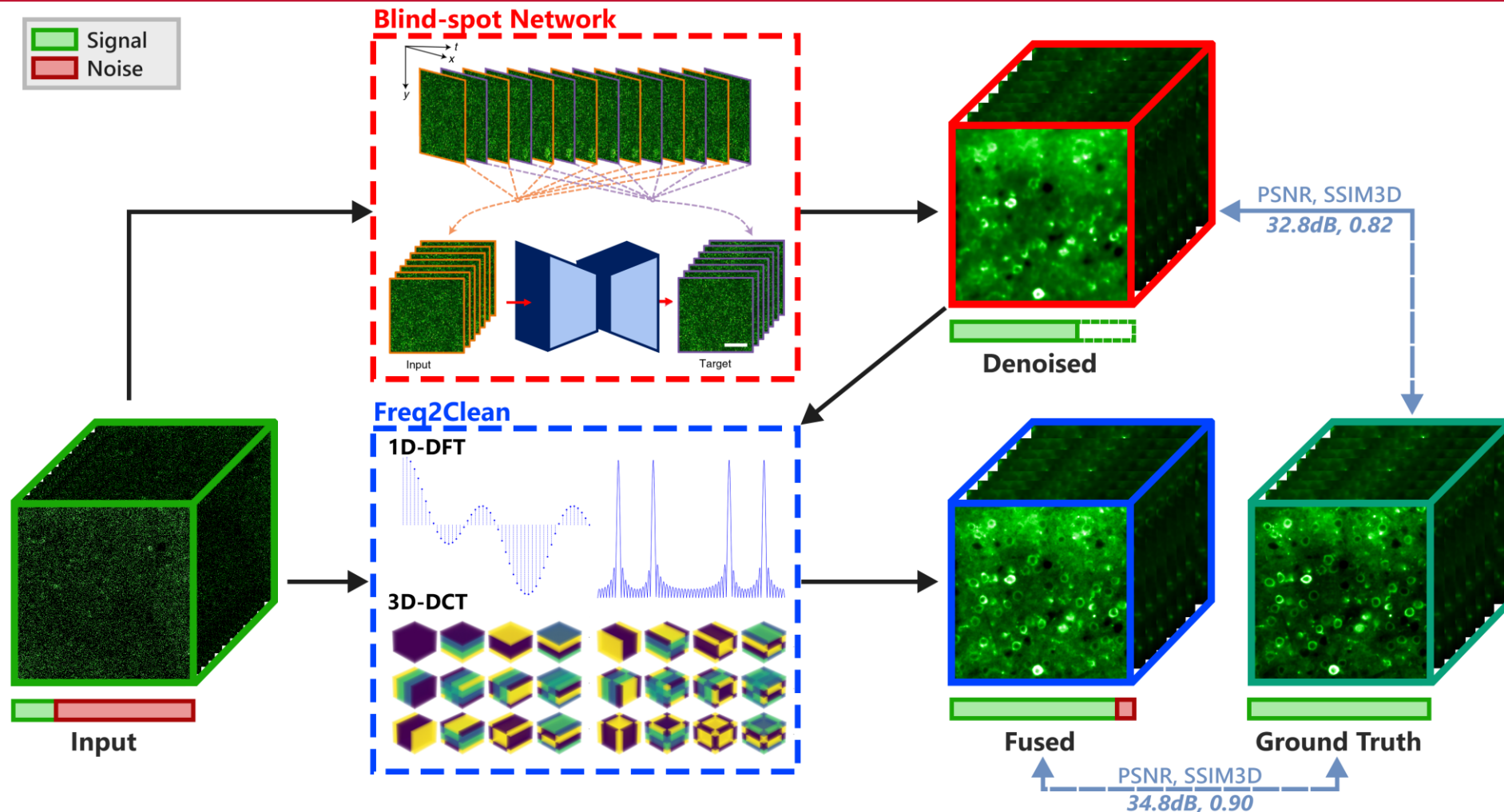
Averaging frames **increases SNR** but **smooths dynamics**.

$$\hat{y} = \text{Poisson}(y) + \mathcal{N}(0, \sigma^2) \Rightarrow SNR = \frac{y}{\sqrt{y + \sigma^2}} \approx \sqrt{y}$$



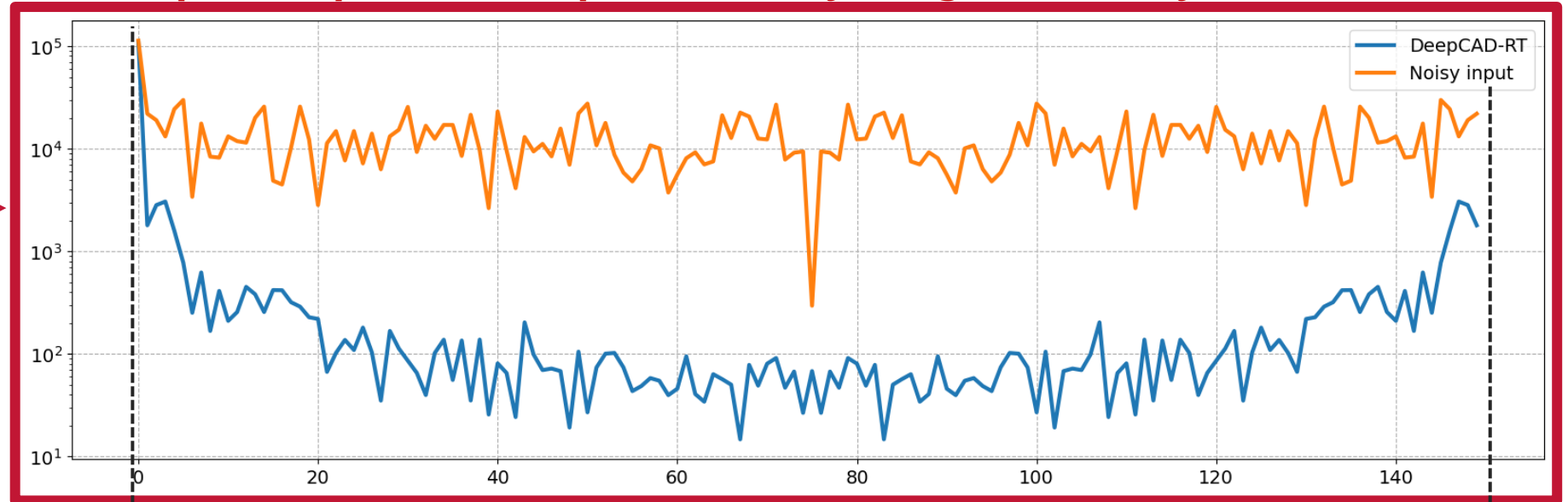
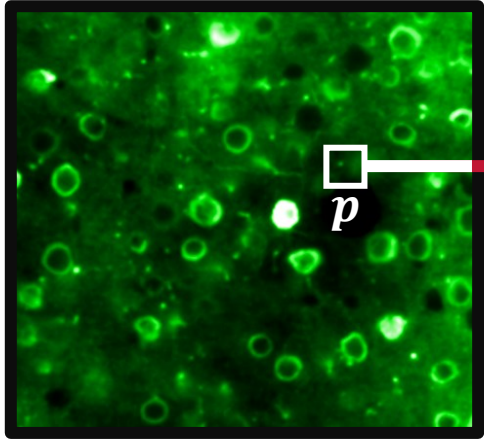
The core idea of our approach

Combine the fine details of averaged frames with the ability of SOTA denoisers to capture fast dynamics. **The drawbacks of each method are offset by the other.**



1D-DFT video fusion: mask parametrization

DFT of pixel p , positive frequencies-only, magnitude-only



$\alpha =$

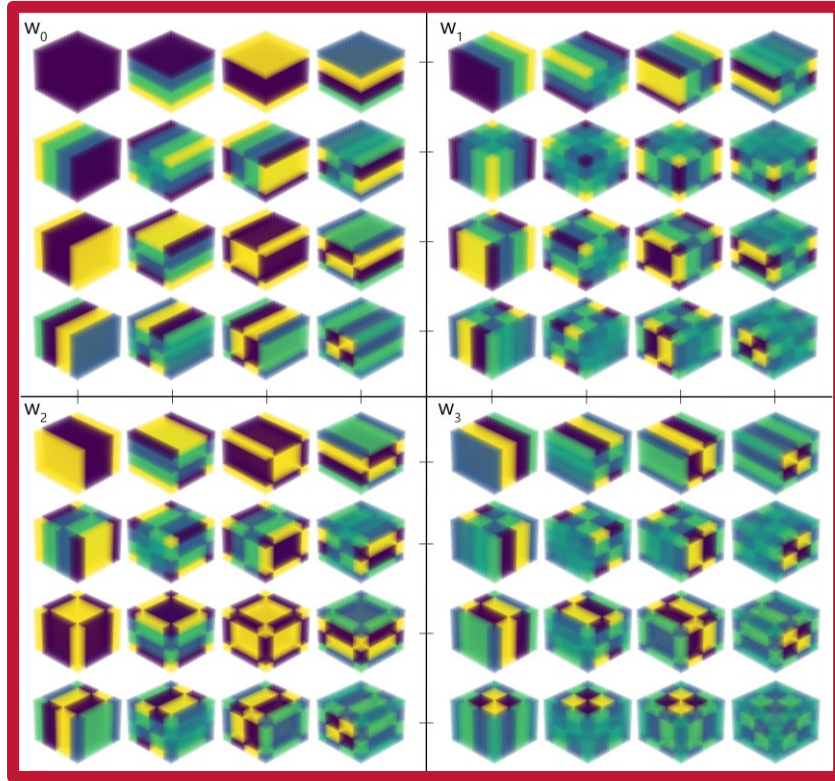
Noisy

DeepCAD

$$|Y_{x,y}|^+ \triangleq \alpha \odot |Y_{x,y}^{noisy}|^+ + \bar{\alpha} \odot |Y_{x,y}^{deepcad}|^+$$

3D-DCT video fusion: mask parametrization

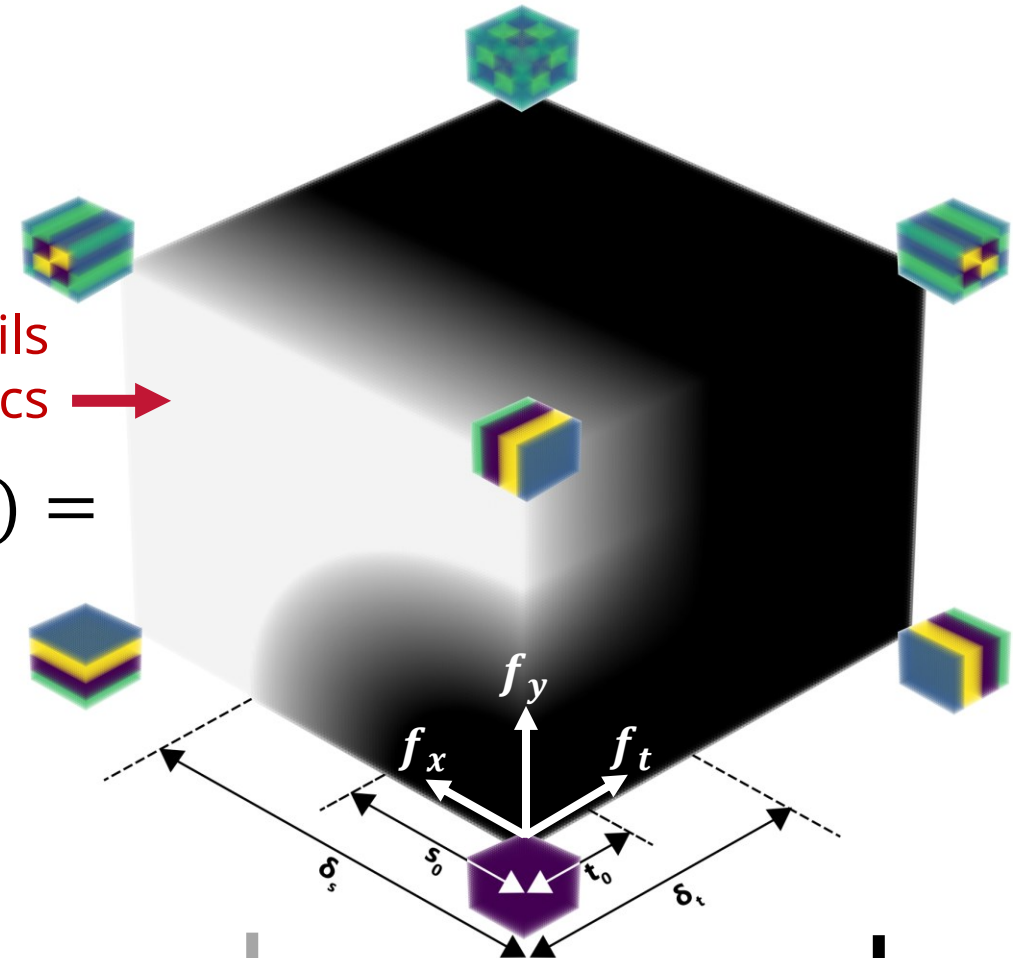
DCT Bases



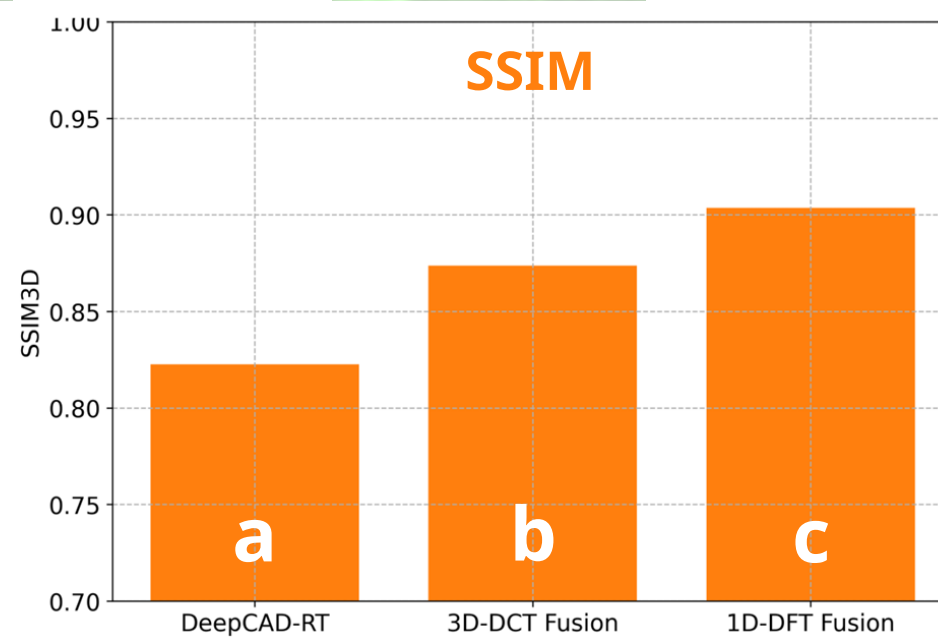
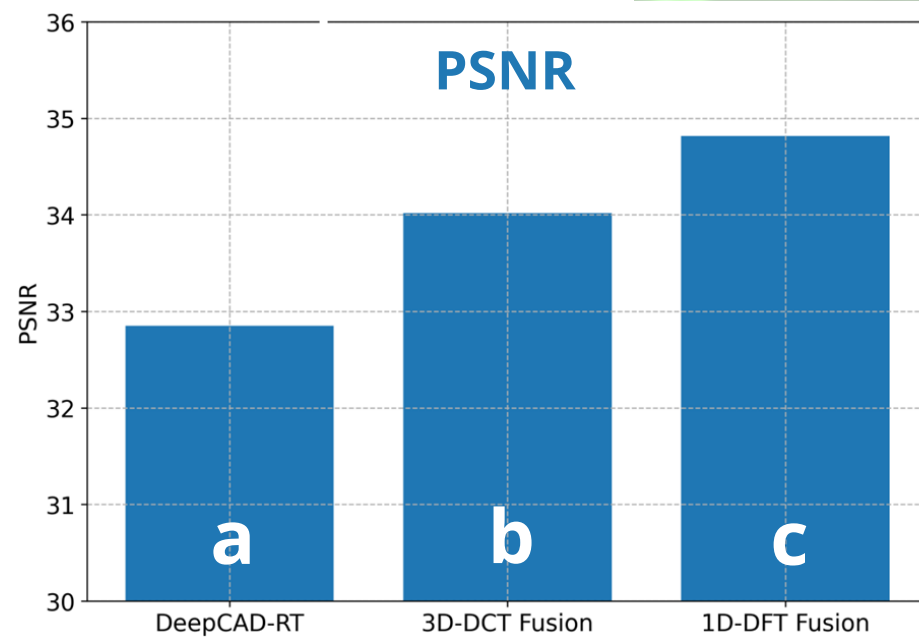
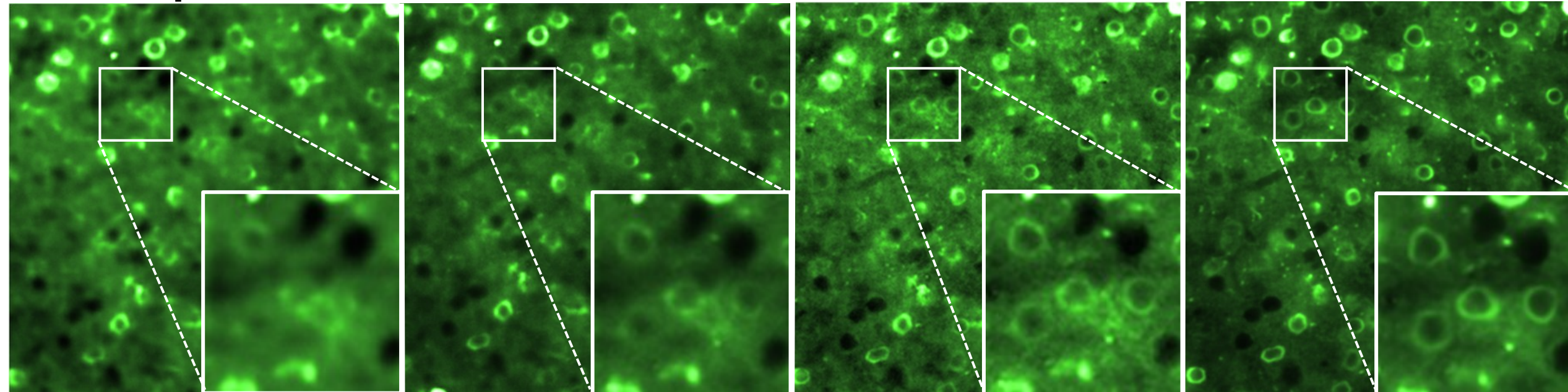
$$y = \alpha_{0,0,0} \times \text{cube}_{0,0,0} + \alpha_{1,0,0} \times \text{cube}_{1,0,0} + \dots + \alpha_{N,N,N} \times \text{cube}_{N,N,N}$$

High-freq details
Low-freq dynamics →

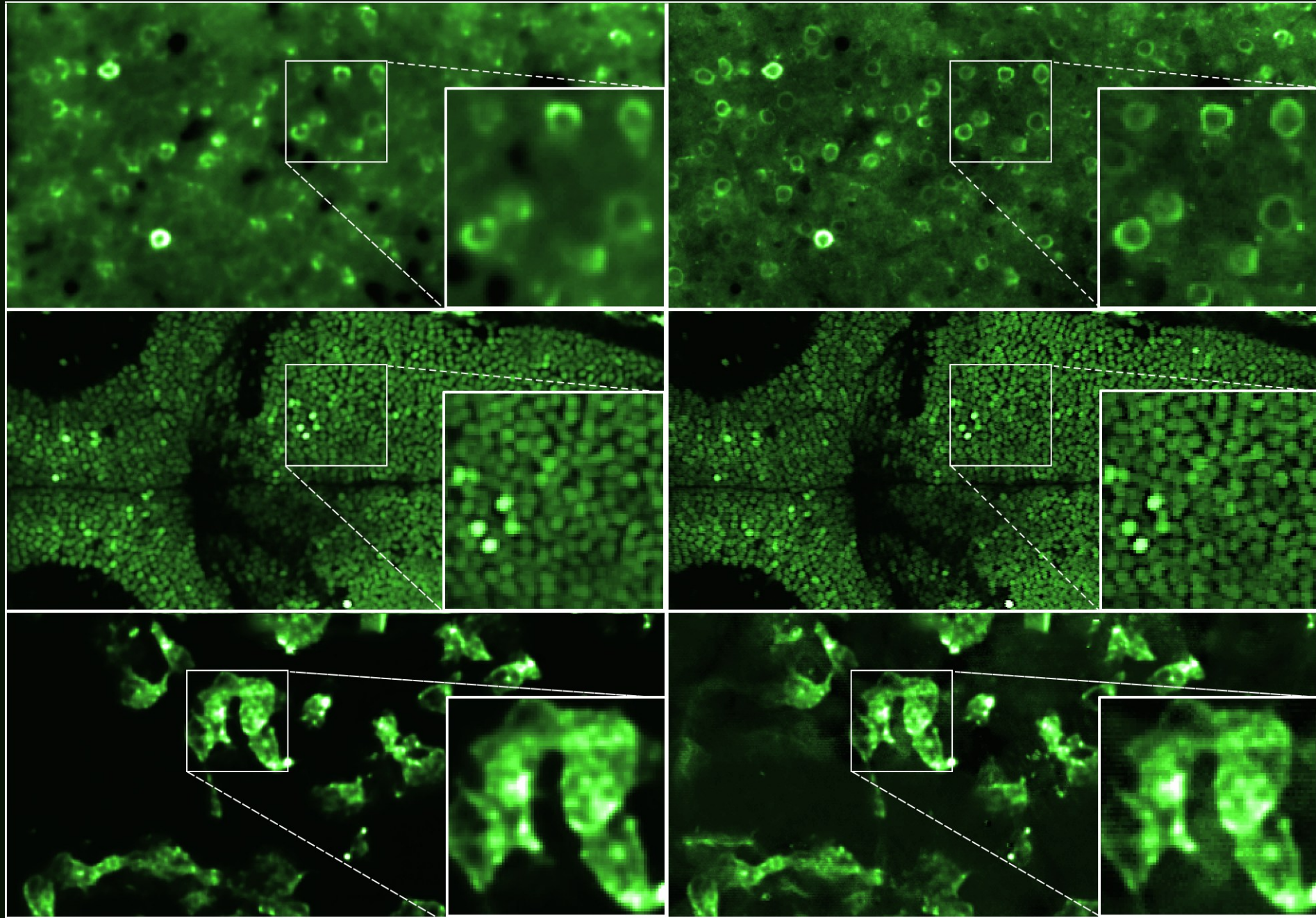
$$M(s_0, \delta_s, t_0, \delta_t) =$$



$$Y \triangleq \boxed{M \odot Y^{avg}} + \bar{M} \odot Y^{deepcad}$$

a) DeepCAD-RT**b) 3D-DCT****c) 1D-DFT****Ground Truth**

DeepCAD-RT vs Freq2Clean

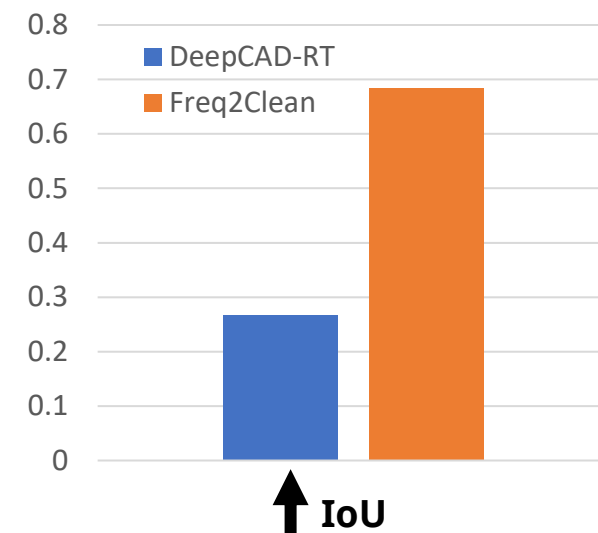
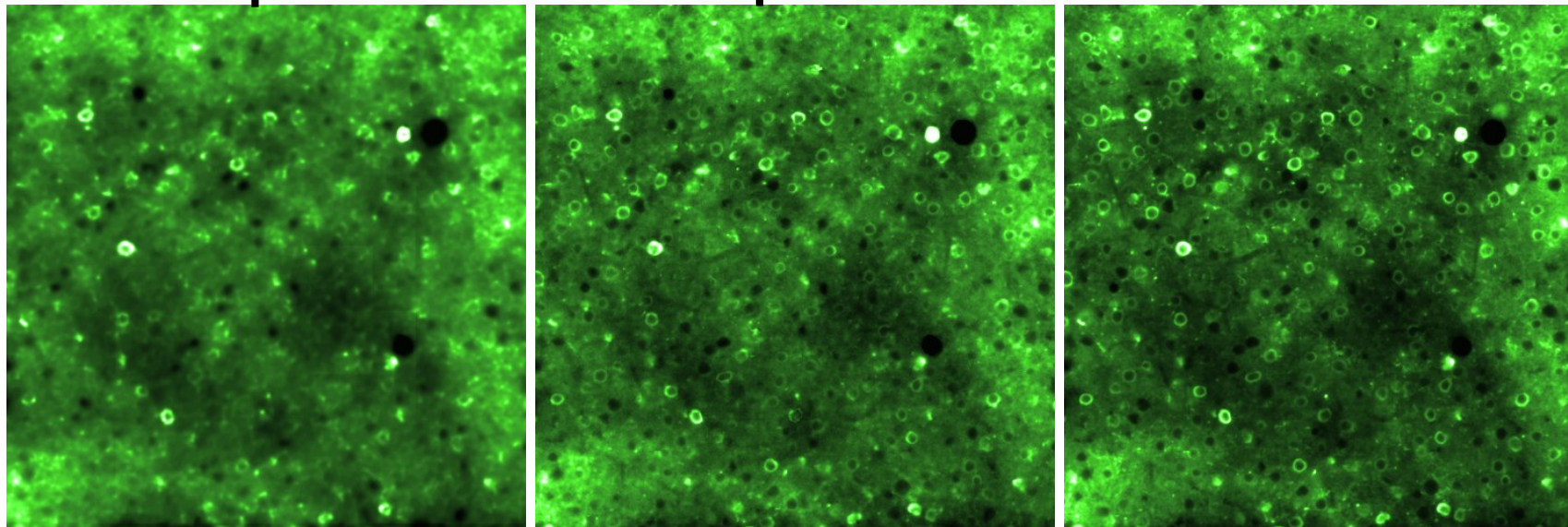


Segmentation improvement

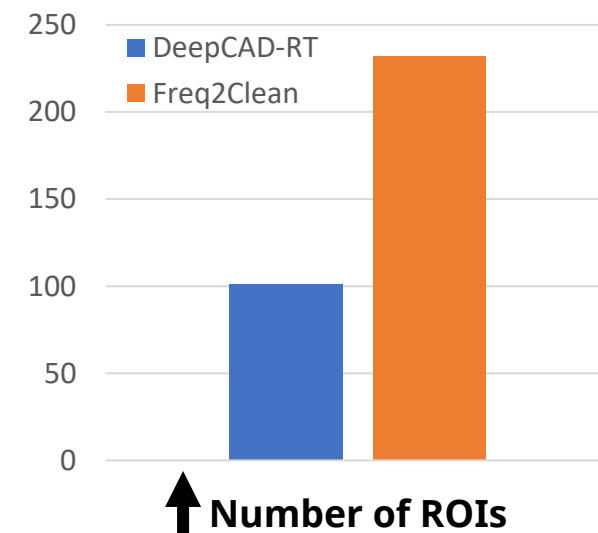
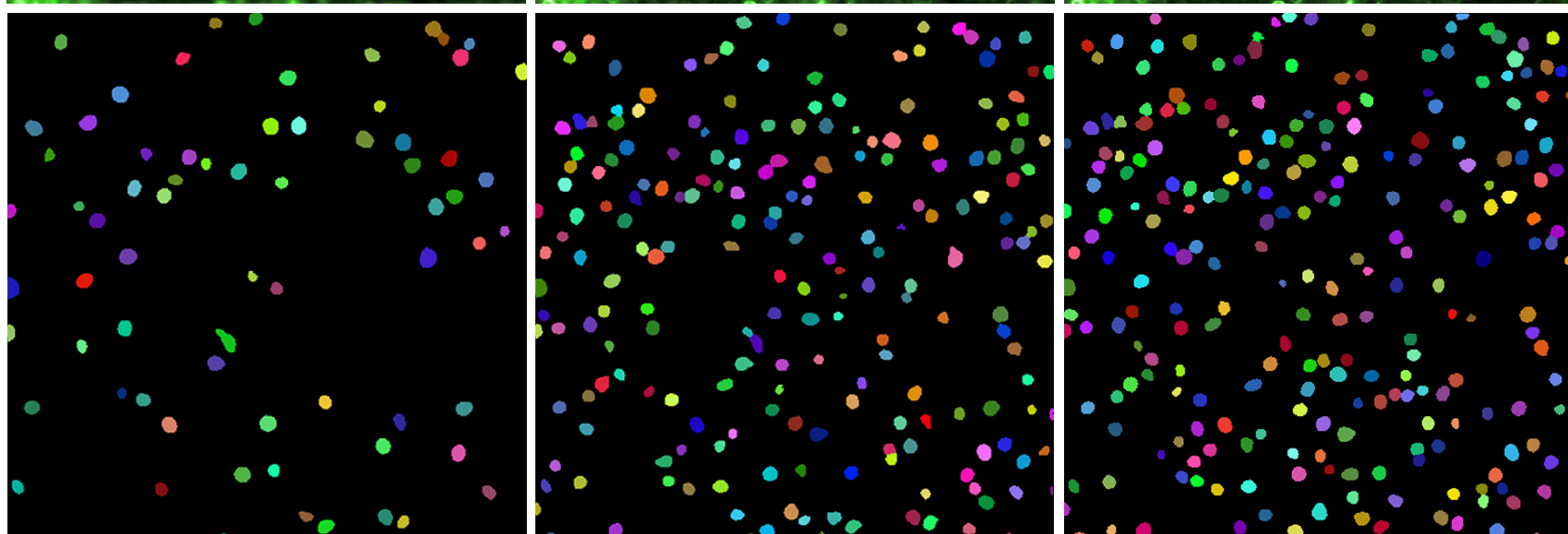
DeepCAD

Freq2Clean

Ground Truth



Cellpose



Conclusion and future developments

Freq2Clean outperforms self-supervised denoisers in terms of **denoising** quality and **segmentation** accuracy, on both synthetic and real-world datasets

Limitations:

- ❑ Noise: **0-mean, weak temporal correlation;**
- ❑ **Absence of spatial dynamics;**
- ❑ Input should have a **very low SNR;**
- ❑ Requires a great number of frames $\Rightarrow$ **no good for real-time scenarios.**

Potentialities:

- ❑ Results hold for **real dataset** and **other self-sup. models;**
- ❑ Can be **extended to other imaging modalities** (*confocal and wide-field fluorescence*);
- ❑ Can handle **4D data** (*fMRI*);
- ❑ Can **handle complex inputs** (*FLIM*);
- ❑ **Phase** is underexplored;
- ❑ Can be **unfolded** into a neural net.



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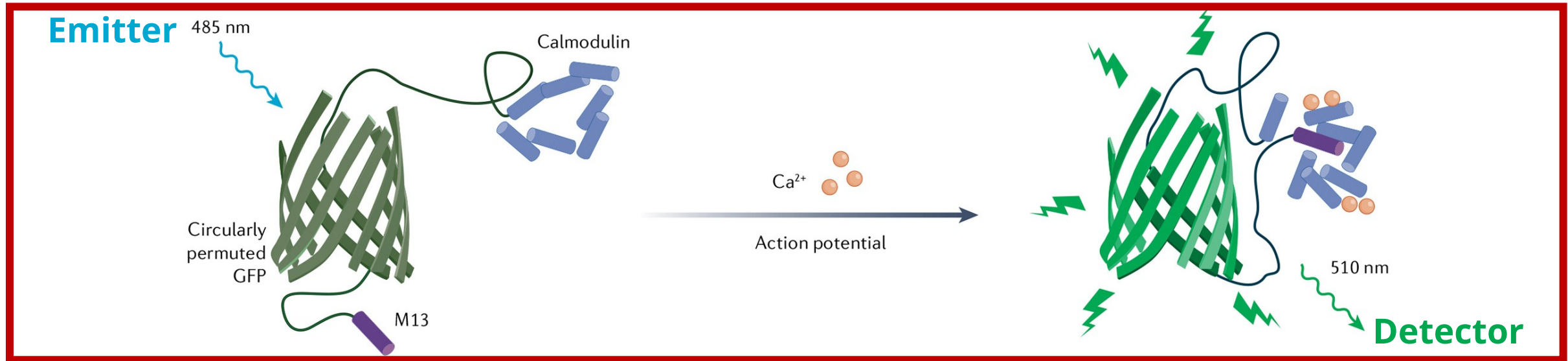
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I sincerely thank Prof. Adriano Mancini, Prof.
Vittorio Murino, Daniele Berardini, PhD, and the
whole Commission.

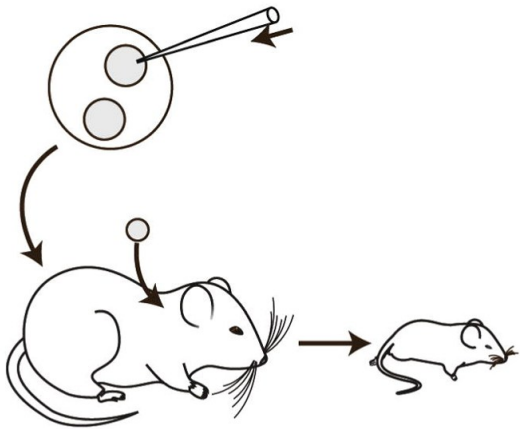
Thank you for your attention

Genetically Encoded Calcium Indicators

GECIs **can target specific cells** when delivered with a promoter. Once a GECI is in the DNA, the cells continually produce the indicator (**chronic imaging**) without repeatedly loading dye.



$$\text{GCaMP6} = \text{GFP} + \text{Calmodulin} + \text{M13}$$



Transgenic mouse lines - Inject DNA into a fertilized mouse egg. The egg is implanted into a surrogate mother, and the offspring that have integrated the DNA into their genome are bred together to create a stable line.

Diffusion models

Deep-Learning networks **implicitly learn image priors** during training. In 2021 it was discovered how to retrieve it and how to sample from it.

$$\nabla_x \log p(x) = \frac{x - D(x, \sigma)}{\sigma^2}$$

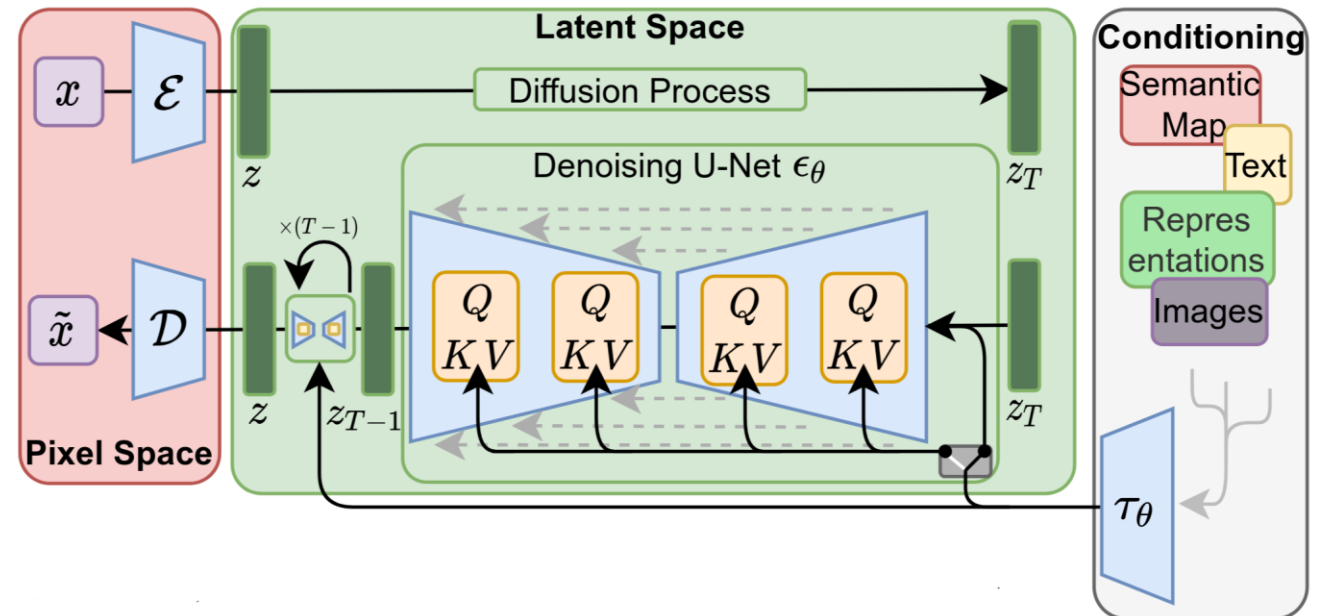
AWGN denoisers **know a lot about priors**

$$\hat{x}_{k+1} = \hat{x}_k + \alpha \nabla_x \log p(\hat{x}_k) + \sqrt{2\alpha} z_k$$

The annealed Langevin dynamics makes **sampling possible through iteration**

However, diffusion models are not well-suited for 2PM denoising:

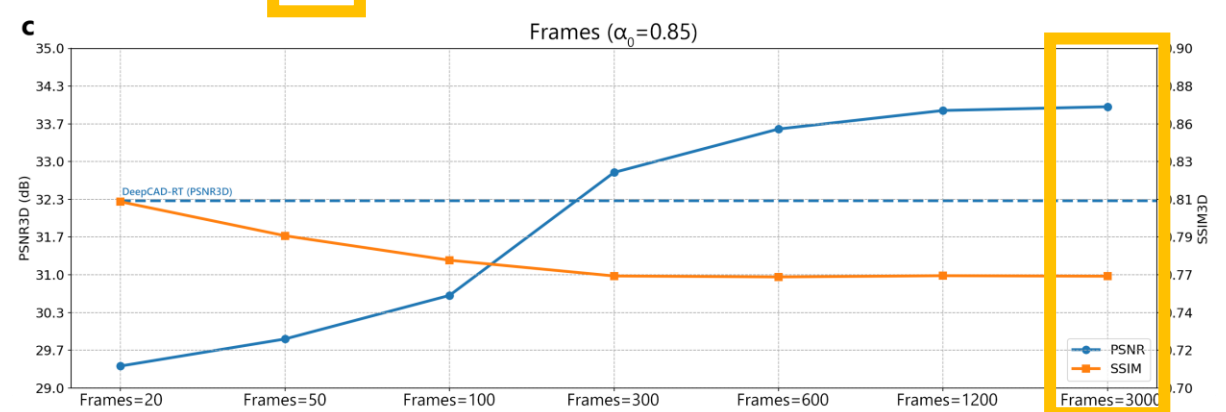
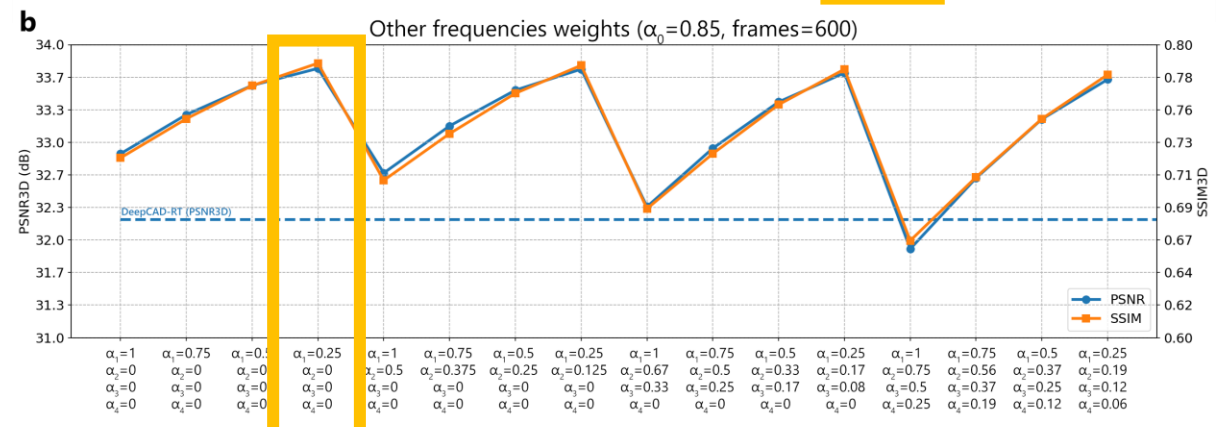
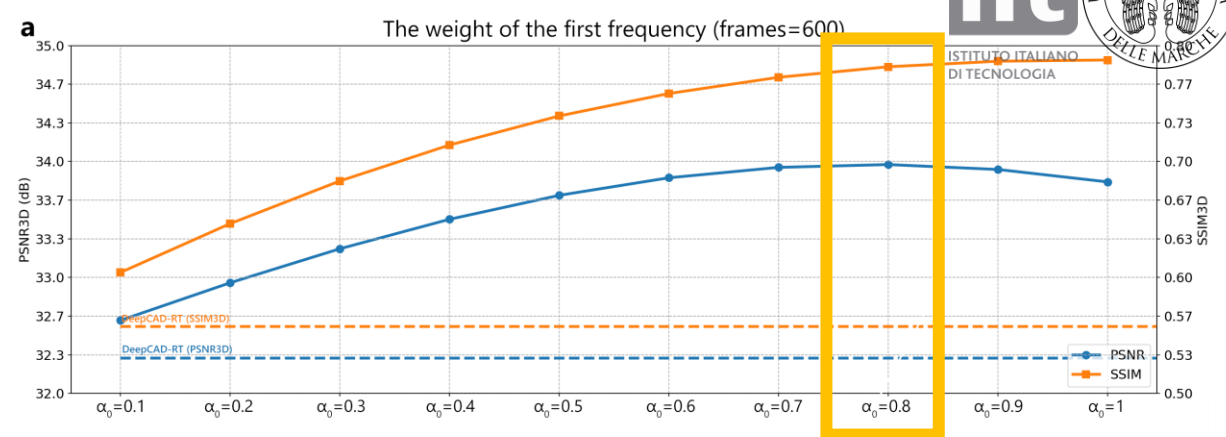
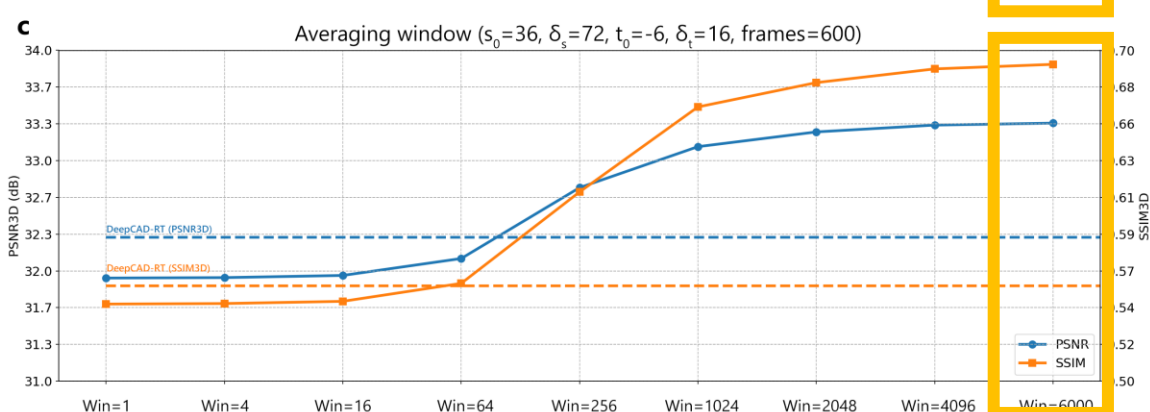
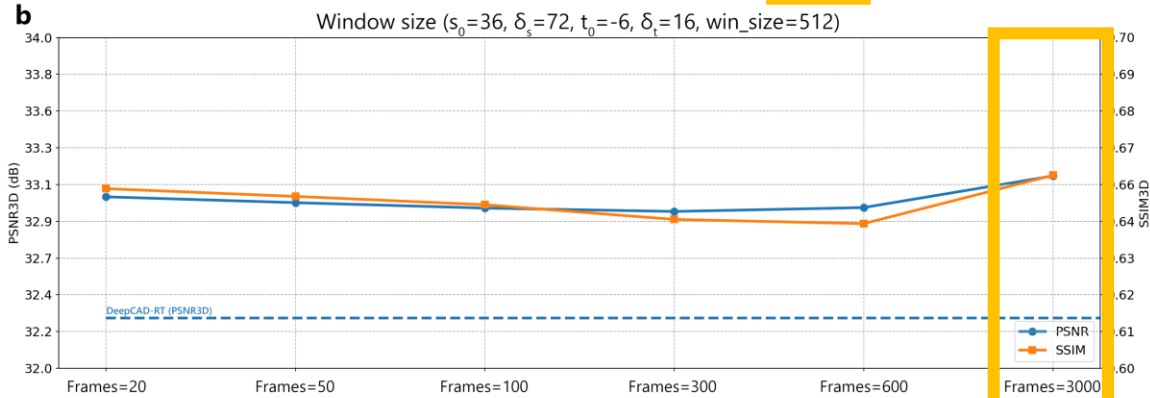
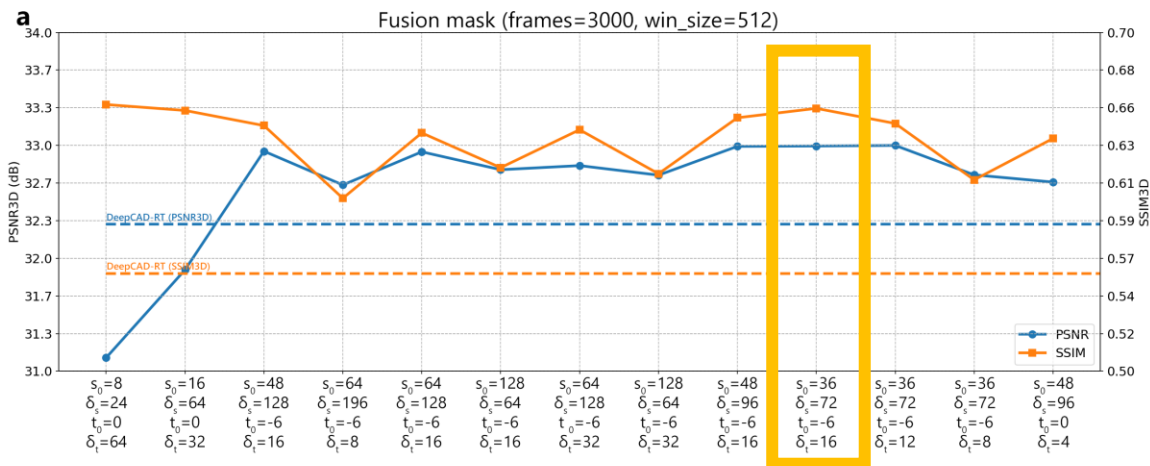
- ❑ **Heavy** training/inference $\Rightarrow$ should be distilled after training
- ❑ **VAEs** must be trained from scratch
- ❑ Often used in **supervised** trainings
- ❑ **Hallucinate** details to deliver crisp images $\Rightarrow$ no good for very low SNRs



3D-DCT

Grid search

1D-DFT



Generalizing the results

➤ Changing denoiser model

Improvements are stable even when selecting other SOTA denoisers as baseline

➤ Changing dataset

Improvements reduce when testing against real datasets:

- ❑ **Pseudo-ground truths** $\Rightarrow$ metrics are approximated
- ❑ **Modest input SNR** $\Rightarrow$ little margin of improvement
- ❑ Neutrophils has **moving cells**

